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Research article

# Transformative climate resilience and sport mega-events – The case of the Australian Open

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ABSTRACT

Adapting to extreme weather events and mitigating greenhouse gases have become critical challenges for sport mega-events. It is clear it is increasingly difficult to host large summer and winter sporting events due to weather extremes such as heat waves, storms or droughts. Extreme heat, in particular, makes events – and event management – less predictable and can have a significant impact on the health of athletes, spectators and staff. International organisations such as the United Nations have called for a rethink, and climate action in sport. At the same time, event organisers are beginning to adapt to the new conditions and develop strategies and measures to reduce event-related greenhouse gases. This paper examines whether sport events can act and function as catalysts for sustainability transitions towards a reduced climate impact. Drawing on the literature on mega-events, urban climate resilience and sustainability transitions, this paper introduces the framework of transformative climate resilience to analyse the climate adaptation and mitigation actions of sport mega-events. Using, as an example, the Australian Open – one of the four largest tennis tournaments in the world – valuable insights are provided into the event's different stakeholders, approaches to climate action, and challenges in building transformative climate resilience.

## 1. Introduction

Increasingly, sport mega-events are challenged by extreme weather events such as heat waves, flash floods, storms and other weather-induced phenomena such as bushfires (e.g. Climate Council 2021; Dingle and Stewart 2018; Scott et al., 2015). The 2021 Summer Olympic Games in Tokyo were one of the hottest in history (The Guardian, 2021). There is little doubt that climate change is, at least partly, responsible (e.g. IPCC 2021). Over the next decades, with expected even higher temperatures, some locations of major winter and summer sport events will likely face serious challenges hosting the events in the way we have known them. While sport events are affected by the outcomes of climate change, they are also a source of greenhouse gas emissions caused by the operating venues, procurement of goods and services, or travel of athletes, event staff and visitors. The initiative Sports for Climate Action Framework, started by the United Nations Framework Convention on Climate Change (UNFCCC), made clear that sport events and sport organisations should more actively transition events towards reduced climate impact, and advocating climate action (UNFCCC, 2021; UNFCCC, 2020). While campaigns addressing societal challenges such as racism and homophobia are well-established initiatives

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in many sports today, calls for climate action are relatively recent. A prominent example is the cooperation between the European Union (EU) and the Union of European Football Associations (UEFA) over the next three years. Through stadium and TV advertising and athletes' initiatives, the goal is to "promote awareness of the climate emergency and inspire more people to take action to protect the environment" (EU/UEFA 2021).

What role sport events could play in building resilience towards extreme weather and driving wider transitions towards more sustainable ways of production and consumption is widely unexplored. Given the cultural importance, and the significant economic and reputational impact of sport events (see e.g. Getz and Page 2016; Ludvigsen et al., 2022; Müller 2015; Preuss 2015), there is a surprising lack of scientific work that links sports with climate resilience and sustainability transitions. In light of recent initiatives, this article aims to contribute to the understanding of how sport mega-events could build transformative climate resilience, understood in this paper along two complementary ways: first, as an adaptive capacity to absorb climate-related stresses; and, second, as a transformative capacity to evolve into more desirable configurations that also accommodate mitigation efforts.

Drawing on the literatures on urban transformative capacity (Castán Broto et al. 2019; Shahani et al., 2022; Wolfram et al., 2019; Wolfram 2016) and literature on the configuring powers of mega-events (Duignan 2021; Glynn 2008; Hayes and Karamichas 2012; Schüßler et al., 2015; Thiel and Grabner 2015) and urban climate resilience (Tyler and Moench 2012), this paper explores how sport events can act as catalysts and intermediary platforms for transformative climate resilience. While there is a strong focus on the role of organisational platforms for building local transformative capacity (such as Urban Transition Labs, Urban Exchanges or Urban Living Labs) (see e.g. Bulkeley et al., 2016, Dickey et al., 2022), sports and, specifically, sporting events are widely ignored in this context. We argue that events like the Australian Open (AO) can act and function as platforms for promoting transformative change. Event organisers can introduce measures that contribute to climate change adaptation and mitigation, and have a wider public reach during the event to promote climate and environmental action, locally and internationally.

Scholars in sustainability transitions have explored how place-specific factors (such as local institutional and policy frameworks, economic pathways, networks, and innovation support structures) enable or hinder transitions in a variety of domains such as energy, transport or building and construction (e.g. Köhler et al. 2019; Binz et al., 2019). Kivimaa et al. (2019) provided a detailed overview of different types of intermediaries in the urban sustainability context. Sport events have been 'blind-spotted' in sustainability transition research, even though the events are part of a billion-dollar industry that gains international attention and are based on distinct socio-technical processes.

This paper sheds light on the AO as a case study. As one of the four Grand Slam tennis tournaments (ahead of the French Open, Wimbledon and US Open), the AO is one of the biggest sport events in the world. Over the last decade, the tournament, which is held annually in January during Australia's summer, has faced a number of direct and indirect weather-related incidents. Heat waves that affected players and spectators, heavy rain, and poor air quality due to bushfire smoke have sparked discussion in the media and amongst experts about how to make the tournament more resilient (The Guardian, 2014). At the same time, environmental organisations are calling for the tournament to not only adapt but actively participate in mitigation efforts (Climate Council 2018, 2021). Based on a qualitative research design that includes semi-structured interviews with AO stakeholders and AO-related reports and documents, the paper analyses how the event has been implementing mitigation and adaptation measures to build transformative climate resilience.

The paper is structured as follows: Section 2 discusses the literature on urban climate resilience and sustainability transition and the potential entry point for sport mega-events as a new element of theoretical discussion. Section 3 discusses the AO as a case study and its broader context. Section 4 outlines the methodological approaches used in this paper. Section 5 presents the research findings, Section 6 discusses the findings, before Section 7 concludes and provides an outlook for further research avenues and implications for practice.

## 2. Sport events as catalyst platforms for transformative climate resilience?

Over the last decade, considerable ambition was expressed by large international organisations. While it was the International Olympic Committee (IOC) that drove the agenda by making the Olympic Games more sustainable, initiatives such as the United Nations (UN) Sports for Climate Action Framework (UNFCCC, 2021) are now creating momentum in the sports world. In 2018, the UN highlighted sport as an important enabler of sustainable development and suggested a stronger recognition of the UN Sustainable Development Goals (UN, 2018). Since then, more than 280 sport organisations have joined the initiative to commit to the Paris Agreement. Sport organisations and clubs such as English Premier League football clubs now report on their climate and sustainability ambitions (UNFCCC, 2021). Sport stars are forming advocacy groups to support climate action (see e.g. Australian football league players for climate change, Australian Football League (AFL) Players for Climate Action (AFLP4CA). In the following section, we outline how transformative climate resilience can be conceptualised and analysed in the context of mega-sport events.

Given the rapidly growing body of research in the areas of sustainability transitions and resilience, it is surprising that the role of sport and, in particular, sport events has hardly been addressed. There is little knowledge about how sports can drive actions directed towards tackling climate, sustainability challenges, and systemic socio-technical change. Given the large popularity of sports (e.g. football, basketball, tennis, baseball), there is little doubt they influence audiences in stadiums and at home. This is specifically the case in the context of sport mega-events such as the FIFA World Cup, the Olympic Games, the Super Bowl, or tournaments such as tennis Grand Slams or Formula 1 grand prix (Horne 2007; Müller 2015; Müller and Gaffney 2018). While aspects of sustainability transitions often remain under the radar in research, a variety of socio-economic outcomes (e.g. Baade and Matheson 2016; Hiller 2000; Müller, 2015; Wolfe et al., 2022), aspects on innovation and learning processes (e.g. Schüßler et al., 2015; Thiel et al. 2015) and urban planning (Preuss 2015; Searle, 2002) have been investigated in mega-events research. Further, literature in Management Studies (e.g. Duignan 2021; Glynn 2008) has highlighted the role of sport mega-events in activating citizen participation. With a few exceptions (e.

g. Caratti and Ferraguto 2012; Dingle and Stewart 2018; Karamichas 2012; Müller et al., 2021; Ross and Orr 2022), much less is known about the role sport events can play in the context of climate change adaptation and mitigation and, more generally, the promoting sustainability transitions. While mega-events have been criticised for various social, environmental and economic impacts and conflicts (Ludvigsen et al., 2022; Müller 2017), there has been no discussion of the potential positive role they can play in promoting sustainability transitions and climate action.

### 2.1. Analysing sport mega-events through the lens of transformative climate resilience

In this paper, we conceptualise sport mega-events as transformative climate resilience platforms by exploring climate resilience and broader sustainability measures at the AO. To better understand the performativity of the identified adaptation and mitigation actions, this study also focuses on the role of stakeholders and sheds light on challenges in building transformative resilience. So far, the catalytic role and impact of mega-events has been studied mainly for infrastructure measures or urban economic development (Müller et al. 2018). The role that events can play as a catalytic platform for broader climate action, resilience and other sustainability-orientated actions is largely unknown.

We introduce the concept of transformative climate resilience by bringing together the two conceptual ideas of 'urban climate resilience' (Tyler and Moench 2012) and 'transformative capacity' (Castán Broto et al. 2019; Shahani et al., 2022; Wolfram 2016; Wolfram et al., 2019). Both concepts are relatively young but gaining popularity in academia, policy and the consultancy world (e.g. Arup 2021). Our literature research using the search term 'urban climate resilience'[1] yielded a relatively low but growing number of 56 documents, with the earliest publication dating back to 2012. The search results show a mix of social and environmental sciences linked to the topic. A more detailed review of the identified literature led to the conclusion that the underlying logic of most of the papers can be linked to the early 'engineering' understandings of resilience. This understanding goes back to the idea of a system's adaptive 'bouncing back' to a stable state after being disturbed by a crisis. The most cited publication, published by Tyler and Moench (2012), emphasises the understanding of resilience as an adaptive process. This logic can be also found in the IPCC's definition of resilience as "the ability of a social, ecological, or socio-ecological system and its components to anticipate, reduce, accommodate, or recover from the effects of a hazardous event or trend in a timely and efficient manner" (IPCC, 2014, p. 1108).

Building resilience against environmental, economic and socio-political uncertainty and risks has become a key agenda in different sectors and at different spatial scales. Despite the focus on adaptation to risks, the understanding of resilience has diversified over the past decade in different fields and scientific communities, for example, urban resilience, economic resilience, or community resilience. The mention of 'resilience' in Sustainable Development Goal 11 (UN, 2015) and major programs such as the Rockefeller Foundation's 100 Resilient Cities (100RC) has elevated the issue of resilience specifically in urban contexts to the world stage and contributed to the advancement of the concept of resilience.

The underlying logic of a reactive 'bouncing back' of systems to a previous state has been expanded to a more proactive 'bouncing forward' that includes moving beyond 'business as usual'. Many urban and regional resilience strategies today can be understood as transformational, reflecting the classic aspect of adaptive capacity to sudden shocks and risks, while seeking to anticipate ongoing stresses and build transformative capacity (e.g. Fastenrath et al., 2019). This has also led to debates about greenhouse gas mitigation to address climate change that go beyond mere adaptation strategies. In the context of climate resilience, Beery (2019) specifically discussed aspects of mitigation by defining climate resilience as "a mix of climate mitigation and climate adaptation that aims to minimise current and future disruptions while promoting opportunities" (Beery 2019, p. 1). This shift in understanding from a reactive bounce-back to more proactive management of challenges is reflected in more recent publications on urban resilience (Elmqvist et al., 2019; Fastenrath and Coenen 2021; Meerow et al., 2016) and regional contexts (Giovannini et al., 2020; Trippl et al., 2023). The most-used definition of urban resilience was articulated by Meerow et al. (2016) who defined urban resilience as the "ability of an urban system – and all its constituent socio-ecological and socio-technical networks across temporal and spatial scales – to maintain or rapidly return to desired functions in the face of a disturbance, to adapt to change, and to quickly transform systems that limit current or future adaptive capacity" (Meerow et al., 2016: 1).

Yet the understanding of resilience as an expression of transformative change is underdeveloped. Approaches are needed that go beyond outcome-solving and move towards a problem-solving matter seeking a more fundamental systemic change towards tackling carbon-intensive and unsustainable practices and technologies. As Carl Folke warned early: "Not only adaptations to current conditions and in the short term, but how to achieve transformations toward more sustainable development pathways is one of the great challenges for humanity in the decades to come" (Folke 2006: 263).

A number of academic debates highlighted the question of how to overcome the focus on adaptation and develop new lenses on intentional change and transformation in different contexts (e.g. O'Brien 2012; Göpfert et al., 2019). An important part of academic debates revolves around institutional capacity-building (Healey 1998) and intermediaries (Hodson et al., 2013; Kivimaa et al., 2019) which are taken up in the recent concept of transformative capacity (Wolfram 2016; Wolfram et al., 2019).

The transformative capacity approach focuses on processes and institutional arrangements that drive or hinder sustainability-orientated endeavours that seek to fundamentally change socio-technical and socio-ecological systems at different spatial scales (Wolfram 2016; Wolfram et al., 2019). This actor- and agency-focused framework looks at the path developments of innovative initiatives within and across urban systems. Transformative change is understood as the result of capacity-building through processes of

[1] Scopus data retrieved in Dec. 2021

social learning and agency. A core assumption is that initiatives and experiments can function as intermediary connections between a wide range of actors and orchestrate socio-technical changes in established urban structures such as transport, building, urban greening or waste management. For transformations in these sectors, it is often government that needs to play an important role, as many processes are defined through urban planning principles. That is why Shahani et al. (2022) distinguish between human agency (building capacity with individuals, networks and organisations) and agency across political-administrative levels, which are both crucial for an effective sustainability innovation. Castán Broto et al. (2019) conclude that the concept of transformative capacity is valuable when it is applied to the evaluation of weaknesses, strengths and opportunities for sustainability goals of a project and experimental approaches. That is why experimental approaches such as living labs, real-world laboratories, knowledge exchanges and other forms are increasingly seen as facilitators and intermediaries for transformative change and future-proofing urban contexts (e.g. Hölscher and Frantzeskaki 2021). Linking the conceptual ideas of urban climate resilience and transformative capacity is an important next step for shaping a framework around transformative climate resilience.

We define transformative climate resilience as the ability of a system or organisation to adapt to climate change-induced phenomena such as extreme weather events in order to continue to function and operate, while simultaneously taking normatively-motivated action for change that focuses on climate mitigation efforts that are transformation-orientated.

To conceptualise transformative climate resilience in the context of mega sport-events, there are some clarifications and specialities to be addressed when it comes to organisational structures. First, while urban experiments are often instigated and orchestrated on a smaller scale, for instance, at the neighbourhood level, mega-events are larger in terms of the number of stakeholders involved and the scale they can impact. By broadcasting events, audiences are larger and can be reached internationally. Therefore, it is necessary to explore inter-scalarity, from local to international, to better understand the spatial dimensions of sport mega-events.

Second, while government agencies are involved in sponsoring, providing infrastructure, marketing and security services, they play a minor role in organising and hosting events. Third, sport mega-events are temporary. While urban experimentations through forms such as living labs are usually designed as ongoing processes, large sport events generally take place for a few weeks.

Based on these considerations and derived from the three strands of literatures introduced above, this study focuses on the following analytical dimensions to gain knowledge about transformative climate resilience in the context of sport mega-events: stakeholders, risk awareness, and climate resilience actions.

#### *2.1.1. Stakeholders*

Following the literature on mega-events and the conceptual approaches of the transformative capacity literature (Wolfram et al., 2019), stakeholders and coalitions are the key for driving change. The stakeholders involved in mega-events are numerous and diverse (Ludvigsen et al., 2022). By identifying stakeholders, this study seeks to understand their key roles and their potential to design and orchestrate climate measures. This approach also offers insights into spatial dimensions, especially the inter-scalarity of sport mega-events. Exploring the sphere of influence of event actors provides important insights into the geography of sustainability transitions (Binz et al., 2019; Coenen et al., 2012).

This distinct lens on stakeholders provides the opportunity to identify stakeholders that are missing and potentially important for setting up a wider transformative climate resilience agenda. Literatures in sustainability transitions and urban resilience have shown that to tackle wicked sustainability challenges, a wide range of stakeholders need to be involved to better define problems and find collaborative solutions (e.g. Fastenrath and Coenen 2021; Kivimaa et al., 2019).

#### *2.1.2. Risk awareness*

Following the resilience literatures, risk awareness is the key to build adaptive capacities. An important part of studying climate resilience is understanding the extent to which stakeholders are aware of the risks and potential shocks that can affect a system or organisation. The urban resilience literature has shown how important a risk analysis is to develop adequate solutions (Meerow et al., 2016). As sport events are impacted more by extreme weather events due to climate change, risk assessment should be conducted as a core element for future event-planning (Ross and Orr 2022). In addition to the increasingly difficult feasibility of outdoor events due to extreme weather phenomena, the focus is certainly on the health of players, spectators and employees.

#### *2.1.3. Climate resilience actions*

The risk analysis leads to risk preparedness for a system or organisation. In the context of sport mega-events, the building of transformative climate resilience requires a holistic approach that considers actions focusing on climate adaptation and mitigation. These actions can be tangible measures such as infrastructures and technologies (Dingle and Stewart 2018), nature-based solutions (Bush and Doyon 2018) or intangible solutions such as organisational aspects, event policies or guidelines (Getz and Page 2016; Horne and Manzenreiter 2006). Further, climate resilience actions can be either adaptive climate action, with a focus on protecting players and spectators against extreme weather conditions, or climate mitigation actions, which focus on reducing the carbon footprint as a normative-driven goal for driving transformative change. Because there are no studies so far on these different types of actions, explorative work is needed.

## 3. The Australian open as a case study

Organised by Tennis Australia, the national tennis association, the Australian Open is held every January as the first of the four Grand Slam tournaments (ahead of the French Open in Paris, Wimbledon in London and US Open in New York). In 2020, before the COVID-19 crisis, the two-week event broke a couple of records: more than 800,000 spectators visited the event on the ground (about

15% overseas visitors); there were 813.8 million broadcast hours in more than 200 countries; and A$387.7 million was added to the Victorian state economy (AO 2020).

Launched in 1905, the AO has had changing host cities besides Melbourne, including Sydney, Adelaide and Brisbane. Since 1973, it has been held in Melbourne, Australia's second-biggest city with a population now of about five million. 1988 was an important turning point: the event was moved from December to January, allowing more players to participate and, more importantly, the current venue, Melbourne Park (formerly Flinders Park), was opened. With strong state government support, the redevelopment of the venue, close to the Central Business District (CBD), including the main stadium, Rod Laver Arena (the first covered Grand Slam arena that allowed playing during extreme weather conditions), was completed.

Since then, AO has become Australia's largest and most internationally recognised sporting event, helping to make Melbourne and other Australian places desirable destinations. The image of the 'Happy Slam' has been shaped by tennis icons such as Roger Federer and other players who have declared the AO their favourite tournament. Players appreciate the summer weather and the atmosphere after the winter break but, also, the short distances to the CBD, which make it more convenient than the other Grand Slams in London, Paris and New York. Left unmentioned is the attractive total prize money (AU$71 million in 2020, winners received AU$4.1 million, see AO 2019).

The positive image of the city is also reflected in a number of city rankings. For more than two decades, Melbourne ranked at the top of rankings on liveable cities. From 2010 to 2017, the city ranked first in *The Economist*'s prestigious ranking. However, the AO 2020 might have been a turning point, as some media outlets reported (e.g. The Guardian, 2020). Spectators and players at the 2020 Australian Open experienced significant rainfall at the beginning of the championship and two consecutive days above 39 °C toward the end of the championship, with poor to hazardous air quality caused by bushfire smoke throughout qualifying rounds and the championship (Terzon, 2020). The organisers' handling of the Australian bushfire crisis and the linked smoke pollution was heavily criticised when players were affected. The fires that hit large areas of Australia's east coast in the summer of 2019/2020 – arguably, Australia's largest natural disaster to date – put Australian climate policy on the world stage (Bott and Fastenrath 2020).

For the next decades, climate models suggest that, under all representative concentration pathways (RCP), there will be an increase in air temperature and wet bulb globe temperature (WBGT) and a decrease in rainfall for Melbourne (Climate Change in Australia, 2018; Newth and Gunasekera, 2018). The next editions of the AO and the actions of the organisers will show whether the reputation of the 'Happy Slam' has been permanently damaged.

## 4. Methods

To analyse and understand the AO's actions and approaches to transformative climate resilience, this study followed a qualitative research design based on triangulation of different primary and secondary sources, bringing them together in a chain of evidence (Yin, 2014). The analysed data included primary interview data, publicly available AO-related reports and other documents, local authority documents, newspaper articles and websites.

To gain a first overview of the case study, the research began with an initial screening and review of publicly available policy and planning documents (), websites (e.g. AO, 2021) and national and international media publications (e.g. ABC News, ESPN, *The Guardian, The New York Times*). This first step was crucial to get a first impression of the link between the AO and issues of climate change, extreme weather events and other sustainability aspects.

Based on these initial insights and the above described theoretical-conceptual entry points for this paper, a guide for a semi-structured expert interview was developed and tested in early 2021. This chosen form of research interviewing leaves space for other perspectives beyond the topics and questions defined in the guide (Flick, 2018). The three analytical dimensions, derived from the conceptual framework, were eventually used for the study and discussed in the interviews: Stakeholders and their scale of influence; Risk awareness; Climate resilience actions.

To discuss the three themes with experts, seven semi-structured in-depth interviews with stakeholders involved in the AO were conducted from March to May 2021. To gain a multi-faceted picture, we sought the views of various experts with direct insight into

| | **Coding keywords** |
|---|---|
| **Stakeholders** (and their scale of influence) | e.g. local government; state government; employees & volunteers; venue; visitors; players; sponsors; emergency services; meteorology services; sport association; media |
| **Risk awareness** | e.g. bushfires; bushfire smoke, drought, flood, heatwaves, storms / strong winds |
| **Climate resilience measures** | Technological/material solutions: e.g. stadium infrastructure |
| | Nature-based solutions: e.g. greening the venue |
| | Organisational aspects, event policies and guidelines: e.g. heat policies |

**Fig. 1.** Coding system for analysis of interview data.

elements of the AO's organisation but are entrusted with different tasks and come from different backgrounds. Interviewees were identified through desktop research or recommendations of other interviewed experts. Each of the participants interviewed was directly involved with the AO, its location or the redevelopment of Melbourne Park. The participants were affiliated with a range of organisations such as local and state government, industry and sporting associations, academia, and consultancy. Amongst those interviewed were representatives of Tennis Australia and the City of Melbourne. The professional fields of the interviewees were linked to the fields of climate change adaptation, health/medicine, sustainability, management, architecture, design management, tourism and events management. Due to the ethical agreements with the interviewees, the affiliations cannot be described in more detail. Therefore, they are anonymised and marked with codes (I01–I07).

With the exception of one written response, all other interviews were conducted online and recorded using Zoom because face-to-face interviews were difficult to conduct due to COVID-19 restrictions. The recordings were transcribed and then shared with the participants for their records and feedback. The transcripts were then analysed with the software NVIVO which enables coding of interview transcripts or other documentation to assist in exploring and analysing text data. The original code list was based on the above three identified themes and was then further adapted and extended during the coding process (see Fig. 1).

In this way, different identified risks, actors and actor groups and their roles, and different types of adaptive and mitigation measures could be more easily allocated and analysed. The semi-structured interview approach was helpful to identify involved stakeholders and better understand processes, vested interests and decision-making. Despite the limited number of seven interviews, we are confident the findings are robust and meaningful. To test the significance and validity of the results, a first draft of this paper was shared with all participating interviewees in August and September 2022 to confirm interview quotes, identify any inaccuracies in the content, and receive feedback.

At the same time and in addition to the interview data, the annual reports of the AO's organiser, Tennis Australia, were reviewed to better understand the organisation's sustainability agendas and actions over the past decades. The publicly available annual reports of Tennis Australia were fully screened and analysed for the period from 2008 to 2021 (Tennis Australia 2022). This step was also helpful to confirm parts of the interview results relating to the AO's climate and sustainability measures.

## 5. Results

This section presents the results of the in-depth analysis of the AO on transformative climate resilience, focusing on the three analytical dimensions.

### 5.1. Stakeholders and their scale of influence

The analysis of stakeholders identified a multiplicity of directly and indirectly involved stakeholders and further identified missing

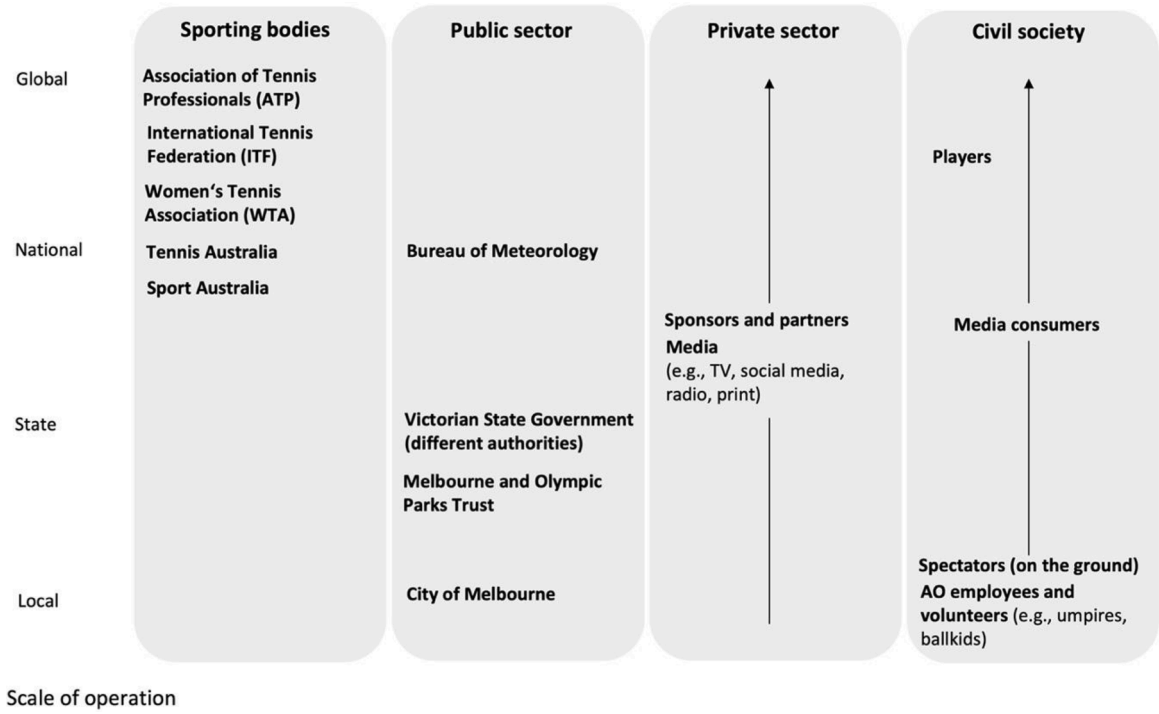


**Fig. 2.** Identified key AO stakeholders (based on interviews and documents) and scale of influence.

ones that would be valuable in driving transformative resilience. This research brings to light a number of actors that operate directly or indirectly and at different spatial levels such as the organising body, Tennis Australia, international tennis sporting bodies, a variety of public sector authorities such as the City of Melbourne and State Government of Victoria, and sponsors, players, spectators and media (see Fig. 2).

#### *5.1.1. Tennis Australia, the organisation behind the Australian Open*

Despite the large number of stakeholders, the organiser, Tennis Australia, must be seen as the most important. Tennis Australia, the governing body for the sport of tennis in Australia, owns and operates the AO. Over the last decades, with the growing popularity and commercialisation of the AO, the organisation has steadily grown, with more than 700 employees and an annual revenue of A$456 million. In June 2019, Tennis Australia announced it was the first Australian sporting organisation to commit to the UN Sports for Climate Change Action Framework (Tennis Australia 2019b). The media release was titled: '"Sustainability is a key focus" as Tennis Australia commits to the United Nations Sports for Climate Change Action Framework' (Tennis Australia 2019b).

An interview partner stated: "Climate leadership and development of mitigation strategies provides opportunities to attract new players, partners and spectators to Melbourne" (I07). Tennis Australia agreed to the goals of the Paris Agreement to reduce climate impact and advocate for climate action by implementing the following five principles: 1) Undertake systematic efforts to promote greater environmental responsibility; 2) Reduce overall climate impact; 3) Educate for climate action; 4) Promote sustainable and responsible consumption; 5) Advocate for climate action through communication (UNFCCC, 2021).

When screening Tennis Australia's annual reports, which are publicly available for the last 13 years, aspects of climate change, sustainability and resilience-building play a minor role. Despite the commitment of the organisation to the UNFCCC in 2019, the reports for 2019/2020 and 2020/2021 (Tennis Australia, 2020), do not provide much information about their efforts regarding the five principles listed above.

The 2019/2020 report only mentions that "a number of sustainability initiatives introduced for AO 2020 included soft plastic diversion, coffee ground diversion and glass recycling" (Tennis Australia, 2020), and that ball kids had worn uniforms made of recycled plastics. Aspects around climate change and resilience-building are not specifically mentioned. While the word 'resilience' appears on page three, the 2020/2021 report also seems empty-handed when it comes to reporting on the company's climate and environmental sustainability efforts. In the 86 pages of the 2020/2021 report, the word climate, is not mentioned once. For the "short- and long-term objectives and strategies" (p. 55) of Tennis Australia, climate action or sustainability aspects are not mentioned.

A search for 'sustainability' resulted in this one sentence: "The ongoing sustainability of clubs, coaches, officials and players was at the heart of the team's thinking as the areas of compliance, safety, communications, advocacy and resourcing formed a specific focus" (Tennis Australia, 2021). Other efforts in promoting environmental sustainability or climate action are absent. An interview partner admits that Tennis Australia is just beginning its sustainability and resilience journey (I07).

In reviewing the descriptions of Tennis Australia's leadership team and board members, it is clear that expertise in sustainability, climate change and resilience-building may be required. Many sport clubs and sport organisations have dedicated sustainability managers, and report on their efforts. Of the 250-plus signatories to the UNFCCC initiative, many have ambitious sustainability and climate action strategies and actions. If the AO seeks to keep up with the promises made when signing the agreement with the UNCCP, new knowledge has to be bundled, and urgent cooperation is required with stakeholders from the public sector (such as departments of health, environment and planning), communities and research. In 2019, Tennis Australia notably explained: "As the national sporting body, we have the power to influence and drive the behavioural changes required to achieve climate neutrality at the local community level" (Tennis Australia 2019b).

As of now, Tennis Australia has not published a sustainability strategy nor has it publicly positioned itself as a supporting platform for exchange in this matter for Australia's tennis clubs and players. Opportunities for more engagement can be identified in the context of information and education for climate change. An interview partner suggested that despite the successful redevelopment of Melbourne Park, something that "could be done much better is using that platform (AO) to build the resilience of the community – the sporting community and spectators – in helping increase their knowledge" (I01).

It will be interesting to observe how the AO will present their climate resilience measures over the next couple of years as the COVID-19 crisis eases. As an interview partner explains, the organisation has now established a sustainability business unit, which was greatly affected by the onset of the pandemic. The focus of this unit lies on "establishing systems for climate change mitigation, and supporting the Paris Agreement (as part of the commitment to the Sports for Climate Action Framework)" (I07).

#### *5.1.2. State government*

The second crucial stakeholder and most important partner for Tennis Australia in hosting the Australian Open is the Victorian State Government. A variety of state government agencies are involved in planning, infrastructure development, marketing, security and licensing. As Section 5.3 will explain in more detail, many of the climate resilience actions and investments have been driven by State Government. This includes the upgrading of stadium infrastructure and other environmental and sustainability initiatives such as waste reduction, energy- and water-efficiency measures and greening of the venue.

In reviewing the AO's key stakeholders through the interview data and document research, we argue that the AO would benefit from stronger connections to local and international climate change experts. Melbourne's internationally recognised researchers and non-government organisations (NGOs) dedicated to climate change would be an ideal complement and support for the AO to work on a climate resilience and sustainability strategy in the coming years.

### 5.2. Risk awareness

The interview data has revealed five key risk areas that are related to extreme weather events: 1) extreme-heat events; 2) droughts; 3) air pollution by bushfire smoke; 4) storms and flash floods; 5) thunderstorms that cause asthma.

Most interview partners see the more frequently occurring extreme-heat events as the biggest risk factor for the AO. During the summer months, particularly Melbourne's inner-city suburbs, experience temperatures of more than 40°. All interviewees agree that there will likely be more, not fewer, days of extreme heat, increasing the vulnerability of athletes and spectators (see also Tennis Australia 2019a).

It is becoming clear that spectators will be more affected during summer outdoor events in Australia, as an interview partner explained: "The Australian Open already has a number of days where the court temperature gets over 50°. And with the number of heat events predicted to rise with climate change, this will only get worse (…) it becomes increasingly difficult to have a safe event for spectators and participants" (I02).

For example, during the 2014 AO, about 1000 visitors were treated for heat-related problems (I04; see also Climate Council 2021). During day time, it's particularly the east side of stands that are heavily affected by heat through direct sun radiation. A health expert for the AO says, "We need to and we have been looking into not just the current but the future, and what we can do to mitigate the risk of heat-related events, and smoke, impact of smoke on sporting performance and health" (I06).

Apart from the risk of heat stress, all interviewed stakeholders (including experts from local and state governments, sport governing bodies, health experts, stadium management) highlighted the impacts of droughts, flood and storms (including rain, dust and asthma). Droughts can potentially affect water supply and increase the chance of bushfires around Melbourne which then might cause serious air pollution as experienced in 2020. Large parts of Melbourne's population were affected, and some AO players had to interrupt their games because of health complaints. Bushfire smoke comes with high levels of fine particles which can cause immediate or long-term problems for the respiratory system (Walter et al., 2020). Gaps in the knowledge associated with bushfire smoke's impact on human health and appropriate mitigation strategies is an issue that requires urgent addressing; interviewee I06 confirms that projects are now established to better understand the impacts on health.

Storms and flash flooding were also mentioned as potential challenges, as the nearby Yarra River could overflow and reach the AO venue. The interviewees also discussed the effect of rain, often resulting in interruption or cancellation of games. Concerns were raised that a potential moving of the event dates to a cooler month could result in an even higher risk of extreme rainfall, and would thus mean a higher chance of cancellations. Thunderstorm asthma events, as Melbourne experienced in 2016 with a large number of fatalities, is also seen as a possibility (e.g. I02, see also Thien et al. 2018).

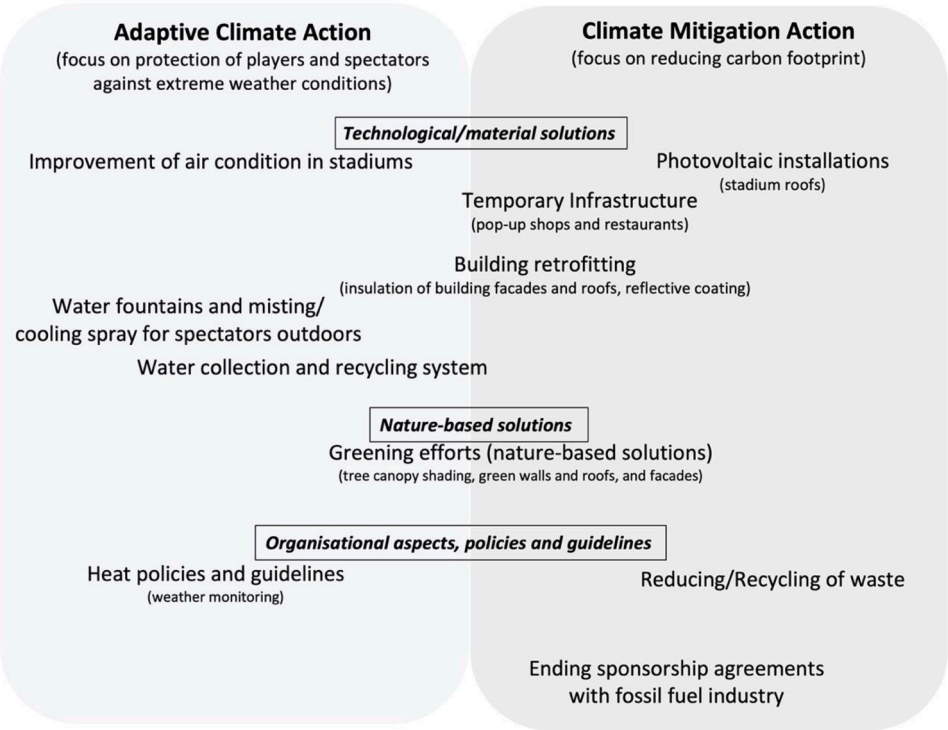


**Fig. 3.** Summary of identified key climate resilience actions in the context of the Australian Open.

### 5.3. Climate resilience measures

Over the last decades, a number of different actions and measures to tackle those risks linked to extreme weather were discussed and implemented by the AO organisers and authorities such as local government and the State of Victoria. The measures span large-scale technological infrastructure solutions, greening initiatives, smaller measures such as drinking water fountains and mist sprayers, and tournament policies based on the analysis of weather data. The variety of actions (see Fig. 3) are explained and discussed in the following chapters.

#### 5.3.1. Technological/material solutions

Many resilience-building actions are targeted at climate adaptation, mostly linked to cooling and shading AO venues. Since 2008, Melbourne Park has undergone a redevelopment to improve facilities. Melbourne Park was built on the former Jolimont Yard, a former train yard located near the CBD, Botanic Gardens and Yarra River. In the 1980s, Melbourne Park merged into a sport and event precinct hosting a variety of sport events, concerts and exhibitions. Responsible for Melbourne Park is the Victorian State Government's Melbourne & Olympic Parks Trust (see VIC Gov 2022). One of the goals of the redevelopment, completed in early 2022, was to build resilience against the increasing likelihood and intensity of extreme weather events, and mitigate emissions (e.g. through increased water and energy efficiency measures). These adaptation and mitigation solutions were not an original goal when first planned in 2008 (I03).

As an act of building water resilience, a water collection and recycling system with a total capacity of 4.5 megalitre (ML) was installed to reduce potable water needs by 60% (see VIC Gov 2022). Water is collected from every hard surface such as roofs, courts, hardscaping, and landscaping on all sorts of asphalt or paved pathways. The recycled water is used for flushing toilets in all new and refurbished buildings, and irrigating Melbourne Park's green spaces (VIC, 2022b[2]).

The latest renovations at the second-largest stadium, Margaret Court Arena, were undertaken by using the US green building standard Leadership in Energy and Environmental Design (LEED) which is slightly above the basic Australian building standard (National Construction Code (NCC)). The rating scheme looking at reduced energy usage and carbon footprint of materials was mentioned by a number of interviewees (I02, I03, I07). Margaret Court Arena, Rod Laver Arena and the Tennis HQ building have achieved LEED gold certification (see also VIC Gov 2022[2]). These efficiency actions included the installation of photovoltaic panels on buildings and special coating on the roof of stadiums (e.g. Margaret Court Arena) that has a high degree of reflectiveness and reduces absorption of heat (I06). However, more ambitious green building standards such as the Australian Green Star six star or LEED Platinum were not considered.

*5.3.1.1. Stadium infrastructure.* The venue's design is at the heart of AO's strategy to build resilience against the impact of extreme weather events. An important measure put in place over the last couple of years was that all spectators, including those with only a general ground pass (limited access to outdoor grounds and minor outdoor matches and training exhibitions), have access to air-conditioned indoor spaces of buildings (I06, I07). Discussions are underway to determine whether more of the 20 courts used for the AO competition should be equipped with shading, ventilation and cooling devices to adapt to extreme weather conditions. These measures would better protect players and spectators from heat-related stress and allow for better event-planning (I01). Currently, only three bigger arenas are facilitated with air-conditioning and retractable roofs, allowing the continuation of the tournament during rain and extreme-heat events. The main arena, Rod Laver, was fitted with an air-conditioning system with multispeed fans to create greater efficiencies compared with its 1980s predecessor (I03). Usually, the top-ranked players have the privilege to play under these conditions. An interviewed expert showed sympathy for the idea of more shaded courts: "Converting them (the courts) to indoor settings, from a climate change adaptation point of view, would be a good thing" (I01).

However, debates about covering the stadiums relate to concerns about the general image of the event. "But from an image perspective, it being an open air, sun, fun event, you know – those images of the crowds being projected around the world of people sitting out there having a (sponsored beverage)" (I05). An interview partner (I07) made clear that there are currently no plans to change the number of indoor courts. Another (I03) explained it is a key event requirement that tennis is played outside and the court is not covered.

*5.3.1.2. Temporary event infrastructure.* During the AO event, a larger area of Melbourne Park's land is used for the construction and operation of temporary infrastructure (such as pop-up shops and restaurants). Most interview partners positively mention the hydration stations and mist available for visitors at various locations. An interview partner (I07) explained these measures are part of the redevelopment plan to reduce waste and costs. The approach of temporary infrastructure thinking is supported by I02, who refers to the limited lifespan of infrastructure, including the need to refurbish infrastructure such as Melbourne Park every 20 years and redevelop every 30–50 years. Over the last few years, temporary buildings and other structures were equipped with outdoor shading at facades or umbrellas (I02).

#### 5.3.2. Nature-based solutions

Interview partners overwhelmingly supported green infrastructure, offering both mitigation and adaptation benefits. New trees, garden beds and lawn areas were planted wherever possible – alignment with the AO image as an outdoor summer event (I02, I04, I07). Green infrastructure appears to be one of the adaptation solutions because it is commonly accepted and bears lower financial costs compared to large infrastructure measures. In addition to adaptation to extreme heat through providing canopy shading and

lower temperatures and providing water storage capacity during floods, green infrastructure has mitigation properties through carbon sequestration (I04). An interview partner made clear that Tennis Australia seeks to work closer with the venue operator, Melbourne & Olympic Parks Trust, to encourage further greening of the site, for year-round use and during the event (I07).

These measures resonate with the current vibrant discussion about nature-based solutions as a non-infrastructure response to a variety of challenges, including the loss of biodiversity, air and noise pollution, and communities' health and wellbeing (Fastenrath et al., 2020; Pineda-Pinto et al., 2022). However, while the greening initiatives in the context of the AO are viewed positively, interviewees articulated opportunities for more greening solutions such as more trees for canopy shading, and infrastructure responses like facades, green walls, green roofs, and soft landing to replace pavement (I01, I02). The challenge and limitation in increasing the number of trees is that large areas are used for temporary stalls, pop-up restaurants and other infrastructure during the AO. I03 explains: "We tried to put in as much (as many trees) as we can, but then the requirements of the event are such that it requires a significant overlay." Another interview partner suggests that more creative solutions should be implemented, and refers to Paris's Grand Slam at Roland-Garros which employs a sustainability manager who puts greening and biodiversity on a higher agenda (I04). There, innovative ideas such as selling honey from the venue's own beehives are sold in shops and aim to help increase awareness of environmental issues.

#### *5.3.3. Organisational aspects, event policies and guidelines*

*5.3.3.1. Extreme-heat policies.* In 2018, as a response to extreme-heat risk, AO introduced its Extreme Heat Policy, which has been updated over the last few years. To protect athletes, a number of weather factors are monitored (radiant heat, humidity, air temperature and wind speed) by the Bureau of Meteorology and onsite to assess risks related to heat and air quality during the AO. More drinking breaks, particularly on the outside courts, and misting fans, which expel cooled water vapour, are now important features at the AO and are part of the climate adaptation solution (I06). In addition, more information about heat is given to AO players and spectators (I06). All interviewees agree that those AO heat policies have been significant improvements. "It's important having policies about this particular temperature, and it's, you know, 50° on court, players should not be out being exposed and playing" (I01).

A health expert explains that the AO heat stress index is based on the 'most vulnerable court', the one that's most at risk. That determines if games are interrupted or suspended (I06). Despite those achievements in terms of policies, stakeholders suggest the organisers could put in place more policies and programs "to make sure events are safe, sustainable, resilient – resilient to what we're likely to see in terms of extreme heat and increasing temperatures" (I01).

I06 suggested the possibility of starting earlier in the morning, which would then allow for a break in the middle of the hottest period of the day. However, that would come with fewer spectators at Melbourne Park and via the TV. Further, more education measures are suggested to better inform players and spectators about the risks of extreme heat.

*5.3.3.2. Changing the event's dates.* A controversial aspect is the possibility of changing the tournament dates. Moving the AO to a later, cooler time of year was discussed with all interviewees as a possible future solution. Some interview partners suggested that starting the event later in the year would positively affect the health of athletes and spectators. An interview partner argued that a later time such as March would likely lower the vulnerability to heat or bushfire smoke (I01).

Not clear is what effect this change would have on the atmosphere, image and the revenue of the AO. Most interview partners point to economic and image constraints that would come with a change of dates. First, advertising revenue from broadcasting would most likely decline because the shift outside of the holiday season would mean fewer television viewers, especially during daytime games. Second, there is a fear visitors, especially domestic, will stay away if the event is moved from the summer holiday period. Third, moving the event out of the school holidays would create challenges for organising many volunteer activities. One interviewee stated that the AO relies heavily on the help of school children as ball kids (I06). Fourth, there is concern about losing the positive image of the AO as a summer event and Melbourne as a destination (I01). An interview partner suggested that "40% of fans that turn up to the Australian Open don't actually watch a tennis game" (I02). Those visitors come for bars and cafes and sponsor activities or to watch the games on big screens on the lawn. A later event would probably affect the numbers of visitors and, therefore, the economic value for hospitality. Many interview partners referred to the image of the AO as the 'Happy Slam'. "It's known as the 'Happy Slam', and it's probably still the 'Happy Slam', but changing the timing of it would likely not necessarily alter its image that much, but it could alter the [economic] value." (I1).

However, one interviewee suggested the heat might turn visitors away in the future. "Do people want to come in and sit by these hot courts at 42° on a regular basis? So, that does impact the commercial side of it" (I05). Fifth, a slight shift forward or backward could be difficult because of the established international tennis calendar and other sporting events in Australia such as cricket around Christmas and the opening of the AFL season in March (I02). An interviewee explained there are no plans to reschedule the event. It is intended to preserve the special nature of the AO as the highlight of the Australian summer (I07).

## 6. Discussion

The findings presented in this paper are novel and offer important insights into the debates whether and to what extent sport mega-events can act as catalysts for transformative climate resilience and sustainability transitions. The analysis of the AO shows that sport mega-events can help drive the implementation of adaptation and mitigation-orientated climate measures. We argue that the focus on events – and, in particular, mega-events in sport – can open up a new academic discussion on resilience and sustainability transitions

because this area has been undervalued and received little attention. In contrast to the much-noted debates on locally initiated experimental platforms through forms such as Urban Transition Labs, Urban Living Labs and Urban Knowledge Exchange as key drivers and bridge-builders (Bulkeley et al., 2016; Dickey et al., 2022), it is argued sport mega-events could have an even larger impact on catalysing transformational capacities due to their local, national and international reach. This also opens up new perspectives around the discussions of urban transformative change (Wolfram et al., 2019) and intermediaries in sustainability transitions, which often focus on the role of government or private sector actors (e.g. Kivimaa et al., 2019).

### 6.1. Stakeholders

It can be argued that sporting events on the scale of the AO, if organised in an inclusive and collaborative way, can act as mediation platforms that pool and translate knowledge on climate and sustainability issues, and create continuous channels for dialogue and discussion. This is often a challenge, however, as sport events have become part of a large and financially-driven industry and are represented by "undemocratic organizations, often with anarchic decision-making and a lack of transparency (…) (Horne and Manzenreiter 2006: 18)". As in other industry sectors, the orientation towards more climate protection requires a multitude of expertise, new actors, and normative goals. To reach the UNFCCC's goals for climate action in the context of sport mega-events, NGOs, community groups, urban planners, policy-makers, sponsors, athletes, spectators, and researchers from natural and social sciences need to be better involved in order to develop effective climate action agendas around sport mega-events. These findings resonate with contemporary urban sustainability and resilience research (e.g. Elmqvist et al., 2019; Meerow et al., 2016). Sport mega-events require further expertise and boundary managers who can intermediate and help to build transformative climate resilience through adaptive and mitigation solutions. The example of the AO shows that climate and sustainability expertise is widely lacking.

### *6.2. Risk awareness*

The study of the AO shows that event stakeholders have a high level of awareness regarding climate change and the associated risks of extreme weather events to the sporting event. The willingness to decarbonise when hosting events, on the other hand, seems to be less pronounced so far. The analysis of the AO unveils that the measures put in place by the organisers are more climate-adaptive than mitigation-focused. This view of climate resilience resonates with the classical understanding of resilience as a responsive 'bouncing back' to a previous systemic state after disturbances (Folke 2006). The idea of 'adaptigation' (Göpfert et al., 2019) has not fully found its way into the event management of the AO. This could change with recent agendas such as those of the UNCCC and other initiatives pushing for sport to play a greater role in climate action.

### *6.3. Climate resilience measures*

The AO has clearly demonstrated how sport mega-events can undertake measures to reduce the impact of extreme weather events to protect athletes, event staff, and spectators. So far, AO's primary focus has been to build climate resilience in relation to the health and wellbeing of athletes and spectators during the event. Efforts in terms of decarbonising the event and advocating for climate change are still undeveloped. Implemented measures are focused on stadium and venue infrastructure and stricter guidelines for extreme weather conditions. We argue that an 'infrastructure fix' is not sufficient to build capacity for a wider shift and response to climate change. The conceptual idea of transformative climate resilience can help researchers and organisers analyse challenges and explore potential climate change mitigation and adaptation measures that can be implemented. These findings are in line with previous studies on the impacts of mega-events (Müller et al. 2018). While material aspects, participation processes and other aspects have been researched, the socio-ecological impacts are largely unexplored.

## 7. Conclusion

The role of sports and sport-mega events, to date, has been blind-spotted in the sustainability transition and resilience research. It is becoming clear that events with global attention such as the Australian Open (AO) are not only affected by the impacts of climate change-induced extreme weather events; they are also contributors to greenhouse gas emissions through aspects such as the operation of venue sites, procurement of goods and services, or travel of athletes, event staff and visitors. International organisations such as the UN have recently suggested sport should use its popularity to promote climate action and implement measures during events to act as a role model (UNFCCC, 2021).

Using the AO as a case study, this paper discusses the role of sport mega-events as catalysts and intermediary platforms for building transformative climate resilience. The conceptual framework of transformative climate resilience is based on literatures on urban (climate) resilience (e.g. Elmqvist et al., 2019) and urban transformative capacity (e.g. Wolfram et al., 2019). The approach helps to analyse and better understand the capacity of sport mega-events to adapt to climate change-related phenomena while taking normatively motivated action for change focused on climate change mitigation efforts. The analysis of the AO shows that sport mega-events can catalyse transformative climate resilience and sustainability issues more generally. In recent years, the AO has introduced a number of tangible meassures, from technological and nature-based solutions and new organisational approaches that have increased the resilience of players and visitors to extreme weather events. Intangible and symbolic measures have also been taken, such as ending sponsorship for companies in the oil and gas industry. Despite these and other aspects, such as increased waste recycling, there is much more that the event can do in terms of decarbonising the event and informing visitors and TV viewers about

climate change. A stronger focus on initiatives that are also publicly effective could unleash transformative dynamics from the local to the global level. However, to become an effective platform driving change in urban socio-technical and socio-ecological systems, new knowledge and transdisciplinary approaches to organising events are needed. Then, sport mega-events can serve as platforms for knowledge translation around urban climate and sustainability issues.

This study leads to new perspectives for future research around sustainability transitions, resilience and sport. A wider interdisciplinary research agenda is pertinent to gain more knowledge about future climate risks, potential adaptive and mitigation solutions, and the challenges of sport events as platforms for building transformative capacity. First, a key aspect will be to shed light on the governance and supporting institutional arrangements around building transformative capacity (e.g. Elmqvist et al., 2019; Raven et al., 2019). This is also linked to the debates around intermediaries in sustainability transitions, which often focus on the role of state and private sector actors (Kivimaa et al., 2019). Second, future research avenues may also include a closer examination of the potential of events to raise awareness and climate advocacy, and educate event visitors and the public about sustainability issues. Third, Ludvigsen et al. (2022) have pointed to controversies and conflicts around sport-mega events. Aspects of greenwashing, lack of social inclusion, just transitions and unintended consequences are further aspects to be addressed in future research on resilience and sustainability of sport mega-events. Particulay the top-down governance making of large events, including fast-tracking laws and corruption, gentrification processes and displacement need exploration (Horne and Manzenreiter, 2006; Ludvigsen et al., 2022). Fourth, research should also look more closely at the geographical aspects of sport mega-events. For example, it is unclear what impact mega-events have on different spatial scales, from local to global. Drawing on the literature on the geography of sustainability transitions (e.g., Binz et al., 2019) would be helpful to better understand the governance of mega-events, their cross-scale interactions and impacts.

Clearly, to date, sport has been ignored as a intermediary platform for climate action and driving sustainability transitions in cities and beyond. While more research is needed, stakeholders involved in mega-sport events should be aware of their potential positive influence and the opportunities to use sport as a catalyst for broader climate awareness and climate adaptation and mitigation action. Adopting the conceptual ideas of transformative climate resilience can help policy-makers and practitioners develop effective strategies and engage the necessary stakeholders.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

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